01 · Product overview

FinPath: AI-assisted financial planning for students and young professionals

A mobile planning layer concept for the RBC Mobile app. It turns transactions into a conservative safe-to-spend figure that reserves upcoming bills and goal savings first, plus budgets that adjust in minutes and savings goals with a monthly target. Every insight shows its calculation.

Business analysis | Requirements | UX/UI design | Working prototype | QA

How to read this case study

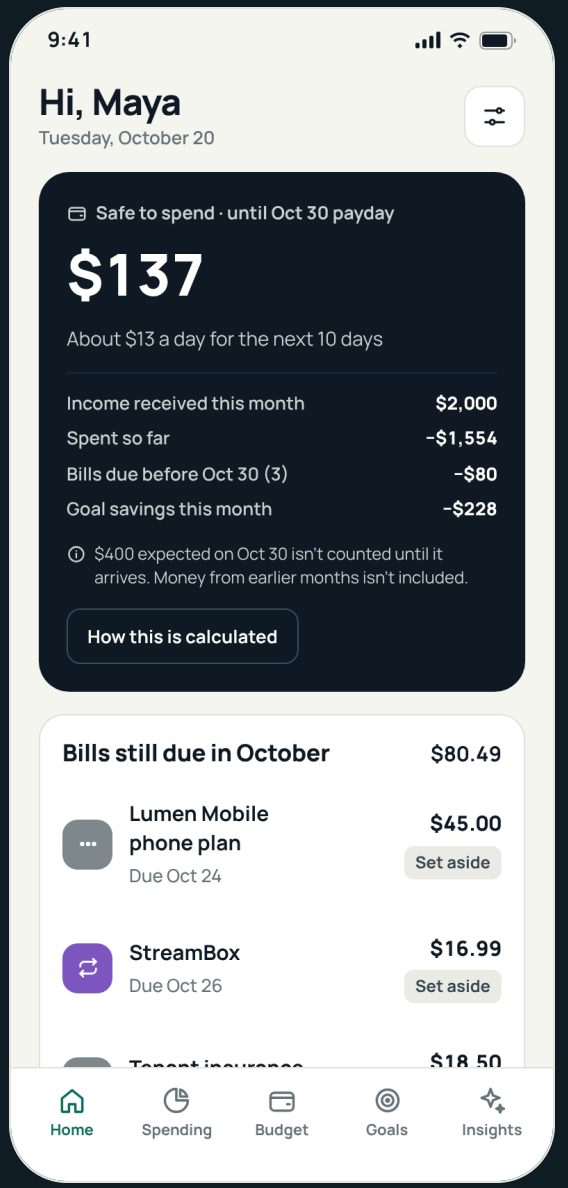
FACT — cited public source

ASSUMPTION — not yet validated with users

PROPOSED — target for a pilot, not a result

IMPLEMENTED — works in the prototype and is tested

No interviews, surveys or usability tests have been run. All prototype data is synthetic. "AI-assisted" insights use labelled rule-based demo logic, not an AI model.



Young Canadians can see their transactions. Turning them into a plan is the hard part.

Banking apps show what was spent. Students and early-career workers still have to work out what they can safely spend before the next deposit, whether a budget is realistic, and how to make steady progress toward a goal.

37% of Gen Z Canadians said they feel in control of their money. FACT

64% of Gen Z reported financial stress several times a week, the highest of any generation. FACT

20% said they don't know where to start managing their money. FACT

TD Bank Group / Edelman Data & Intelligence, Harris Poll Omni, fielded July 28–29, 2025; nationally representative sample of 3,037 Canadians, about 500 aged 18–28. Released October 14, 2025.

Problem statement

Students and young professionals lack a simple, trustworthy way to turn spending into a plan they can act on daily. This plausibly contributes to the low confidence and frequent stress in the survey data. Whether FinPath's approach reduces it is the central hypothesis to test.

ASSUMPTION A-1 A-3

Opportunity assessment

FACTOR	ASSESSMENT
Need	Strong public signal on stress and low confidence (above). Student-specific needs, such as irregular income, are assumptions (A-1, A-5).
Existing tools	RBC NOMI already covers spending insights, suggested budgets, a 7-day payment forecast and automated saving. FinPath has to add to these, not repeat them (page 4).
Timing	Consumer-Driven Banking Act: royal assent March 2026; draft regulations of June 27, 2026 set phase-one, read-only data sharing. The MVP does not depend on it.
Risk	Privacy expectations, explainability, and the line between education and advice.

Sources: newswire.ca (TD, Oct 14 2025) · Canada Gazette Part I, Vol. 160, No. 26 (June 27 2026)

Three life stages, one need: a plan that fits irregular money

Proto-personas are assumption-based. Names and details are illustrative, written to make assumptions testable, not drawn from interviews.

PROTO-PERSONA

Maya, 20 · University student

Part-time job plus term-based student aid; shares rent with roommates.

Job to be done: "Make my money last until the next deposit."

Likely pain: fixed monthly budgets break when income arrives in lumps.

A-1 A-5 A-8

PROTO-PERSONA

Jordan, 23 · New graduate

Steady salary for the first time; student loan repayment starting.

Job to be done: "Set up a budget that sticks and start an emergency fund."

Likely pain: blank budgeting templates feel like homework.

A-3 A-4 A-7

PROTO-PERSONA

Priya, 27 · Young professional

Higher income, more fixed costs, competing goals such as a down payment.

Job to be done: "See if I'm on track without building spreadsheets."

Likely pain: no single view of goals against spending.

A-4 A-9

Riskiest assumptions to validate first

ID	ASSUMPTION	VALIDATION METHOD
A-1	Users want guidance on what to do next, not only a record of spending	8–10 problem interviews; fake-door test of an insight card
A-2	Users will trust bank-built analysis if data use is explained plainly	Concept test of privacy screen; insight opt-in rate in pilot
A-3	A safe-to-spend number is easier to act on than a full budget table	A/B test two dashboard variants on "Can I afford this?"

Full register of 10 assumptions in the Product Analysis Pack. Life stage is captured in onboarding but does not yet change the prototype experience. 3 / 12

NOMI already does much of this. FinPath must earn its place.

Value proposition. For Canadian students and young professionals, FinPath turns transactions into one conservative **safe-to-spend** figure, a **budget** they can adjust in minutes, and **goals** with a monthly amount. Every insight shows its numbers and rule.

CAPABILITY	RBC NOMI (AS DOCUMENTED IN SOURCES)	FINPATH CONCEPT (PROTOTYPE)	ASSESSMENT
Spending insights	FACT NOMI Insights analyses monthly cash flow, categorises spending and shows "where your money is going and what's coming up."	IMPLEMENTED Category chart with filters; 5 rule-based insight types, each with its numbers and rule.	**Overlap.** Showing the calculation is FinPath's proposed difference; whether NOMI explains its insights is not described in the sources reviewed.
Budgets	FACT NOMI Budgets (2019) recommends budgets from spending history, lets clients adjust them, and notifies at 50%, 75% and 100%.	IMPLEMENTED Limits pre-filled from last month for 7 categories, all editable; in-app warning at 80% and over-budget state. No notifications.	**Overlap; NOMI ahead** on proactive notifications. Category lists differ; NOMI's current categories may have changed since 2019.
Upcoming payments	FACT NOMI Forecast (2021): 7-day view of upcoming pre-authorised payments, later extended to bill payments, e-transfers, investment contributions and salary. 900,000+ clients by April 2023.	IMPLEMENTED Bills still due this month and expected income feed one safe-to-spend figure (synthetic schedule).	**Strong overlap in data.** FinPath's bet is the combination: bills, income timing and goal savings in one number. A real build should reuse Forecast's predictions.
Automated saving	FACT NOMI Find & Save sets aside amounts clients are unlikely to miss; clients had saved $83M+ by February 2019.	Not offered. Goal contributions are logged by the user; no money moves.	**NOMI ahead.** FinPath does not compete here and could point goals at Find & Save.
Named goals with a date	Not described in the sources reviewed.	IMPLEMENTED Name, target, month, monthly amount (BR-4), contributions, completion.	**Unverified.** Not a confirmed gap: check the current RBC Mobile app before claiming one.
Questions	FACT Ask NOMI answers questions about recent transactions, helps cancel an e-Transfer and guides clients around the app.	Ask FinPath: keyword matching on synthetic data. Not AI.	**NOMI ahead**: live, on real data.

Positioning

An extension inside RBC Mobile that builds on NOMI's data (categories, Forecast predictions, Find & Save), not a competing app. The case for building it depends on three bets.

Differentiation bets UNVALIDATED

B1 One conservative number that reserves bills, income timing and goal savings is easier to act on (A-3). **B2** Goal-first planning motivates saving (A-4). **B3** Showing the math builds trust (A-2).

Sources: rbcroyalbank.com/mobile/feature/nomi · RBC releases Apr 25 2019, Sep 23 2021, Apr 28 2023 · RBC "What is NOMI Find & Save"

Clear rules first, so design and build agree on the numbers

Business objectives TARGETS TO CONFIRM

- Raise financial confidence for pilot users aged 18–30 (self-reported, 60 days).
- Help new users set up a plan in their first session.
- Build habitual engagement with digital banking (month-2 activity).
- Maintain trust (insight opt-in rate, privacy complaints).

In MVP scope

- Onboarding, privacy choices
- Safe to spend
- Spending breakdown, search
- Editable budgets
- Savings goals
- Explained insights
- Settings, data deletion

Out of scope

- Moving money or payments
- Investment, credit or tax advice
- Accounts at other institutions
- Live AI chat

Business rules and implementation status

ID	RULE	STATUS
BR-1	**Revised in v2.** Safe to spend = income already received this month − spending to date − bills still due before the next expected income − this month's goal savings. Expected income is not counted until it arrives. Rounded down; $0 with a reason when negative.	IMPLEMENTED, TESTED
BR-2	Budget limit is whole dollars from $0 to $10,000.	IMPLEMENTED, TESTED
BR-3	Budgeting more than planned income triggers a warning, not a block.	IMPLEMENTED, TESTED
BR-4	Goal monthly amount = (target − saved) ÷ months from this month up to the target month, rounded up; at least 1 month.	IMPLEMENTED, TESTED
BR-5	Insights educate and nudge; they never recommend financial products.	CONTENT RULE met by current wording; not enforced in code
BR-6	Optional data uses are off until the user turns them on.	IMPLEMENTED, TESTED switches only; see page 11
BR-7	Users can delete all data at any time, after confirmation.	IMPLEMENTED, TESTED

16 functional and 8 non-functional requirements are in the Product Analysis Pack. NFR-1's 44 px touch-target goal is not yet met (page 11).

Maya's first month, from sceptical to in control

Journey and service blueprint for the student proto-persona. Feelings are hypothesised, not observed; the bottom rows show where FinPath's design responds.

STAGE	Discover	Set up	First look	Plan	Live with it	Review
GOAL	Find out if it can help	Start without effort	See where money goes	Make a budget and a goal	Stay on track daily	See progress, adjust
ACTIONS	Sees entry point in banking app	Name, life stage, income; privacy choices	Reads dashboard; taps biggest category	Edits limits; creates emergency fund	Checks safe to spend before buying	Reads insights; adds a contribution
FEELING (ASSUMED)	Curious, sceptical	Cautious about data	Surprised by delivery total	In control	Reassured, or anxious if over	Motivated
FRONTSTAGE	Home banner	Onboarding 1–3	Dashboard, Spending	Budget, Goals	Dashboard, bills card	Insights, Goals
BACKSTAGE	Eligibility check	Consent stored	Categorisation	Pre-fill from last month	Bill and income schedule (could reuse NOMI Forecast)	Monthly insight rules
PAIN TO WATCH	Unclear benefit	Too many questions	Wrong categories	Blank-page budgeting	Irregular income; bills due before payday (A-5)	Generic advice
DESIGN RESPONSE	One-line value proposition	Two short steps, or sample data	Tap category to filter	Suggested limits	Safe to spend counts only money received, after bills and goals	"How we calculated this"

The MVP tests the riskiest assumptions with the least build

Must

- **US-1** Set up in a few steps
- **US-2** Choose data uses
- **US-3** Know what's safe to spend before my next income
- **US-4** See top categories
- **US-5** Set category limits
- **US-6** Warn before overspending
- **US-7** Create a goal with a date
- **US-8** Log contributions
- **US-9** Insights that explain why
- **US-10** Clear loading and error feedback
- **US-12** Delete my data

Should

- **US-13** Larger text

Search (FR-6), daily chart (FR-5) and goal deletion (FR-11) are also Should-level requirements.

Could

- **US-11** Ask a quick question about spending
- **US-14** Monthly check-in reminder

Won't (now)

- **US-15** Include accounts at other banks

Depends on open banking phase one. Also deferred: automatic transfers, recategorising, shared budgets, term-based budgets, ML insights.

Example acceptance criteria · US-3 Safe to spend before my next income

Given income has been received and a deposit is still expected this month, the dashboard shows safe to spend until that payday: income received, minus spending to date, minus bills due before payday, minus this month's goal savings, rounded down. The expected deposit is named and excluded. When the result is negative it shows $0 and the shortfall. `IMPLEMENTED AND TESTED`

A quiet system where colour only ever means something

Design principles

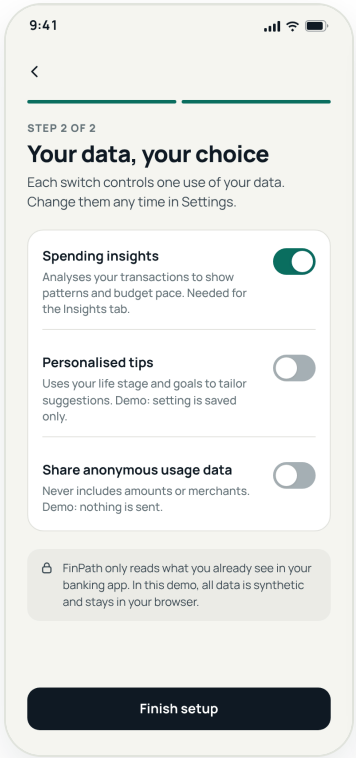
- **One number first, and a cautious one.** Safe to spend leads the dashboard and never counts money that hasn't arrived.
- **Show the math.** Safe to spend and every insight open their numbers and rule.
- **State in words, not just colour.** Budget status always has a label and icon.
- **Respect the user.** No guilt language; optional data uses start off.

System

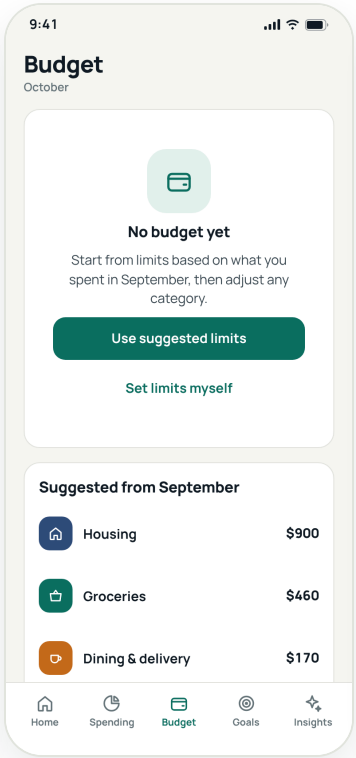
- Ink and canvas for structure; one teal accent for actions and progress.
- Green, amber, red reserved for budget state.
- Manrope with tabular figures for amounts.
- 17 component specs with defined states; 4-point spacing.

States designed

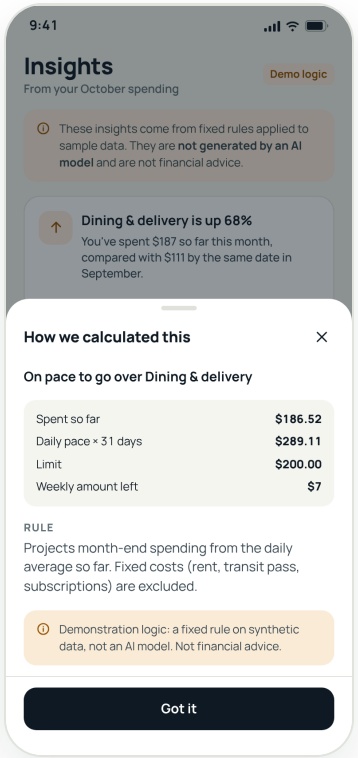
Empty (new user, no goals, no search results), loading (skeletons), error (connection failure with retry), validation errors, over plan with the shortfall, goal complete, confirmation dialogs. 22 frames in the screen inventory.



Privacy choices, optional uses off



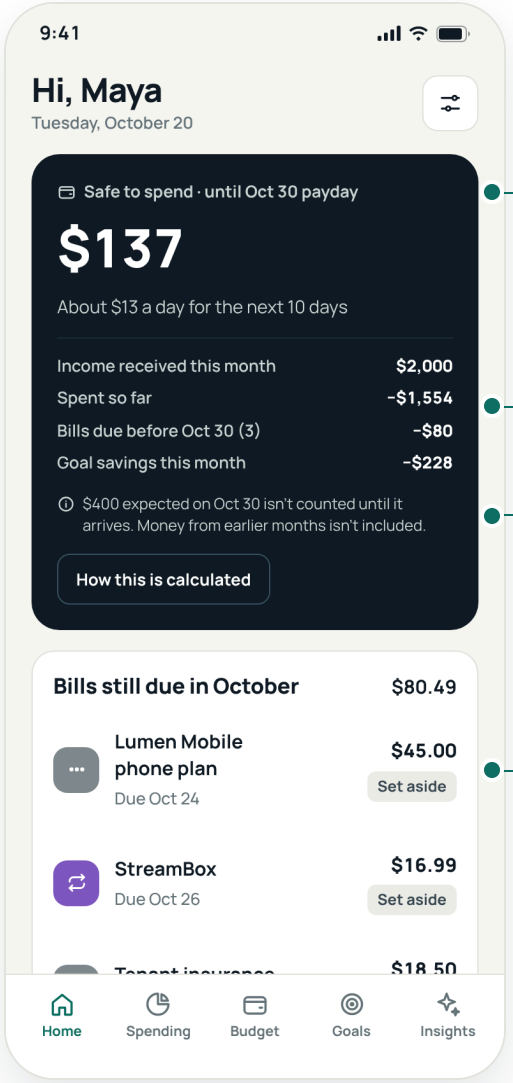
Empty state with a next step



Every insight shows its numbers and rule

Screens are rendered by the prototype's own code

The two screens that carry the product, annotated



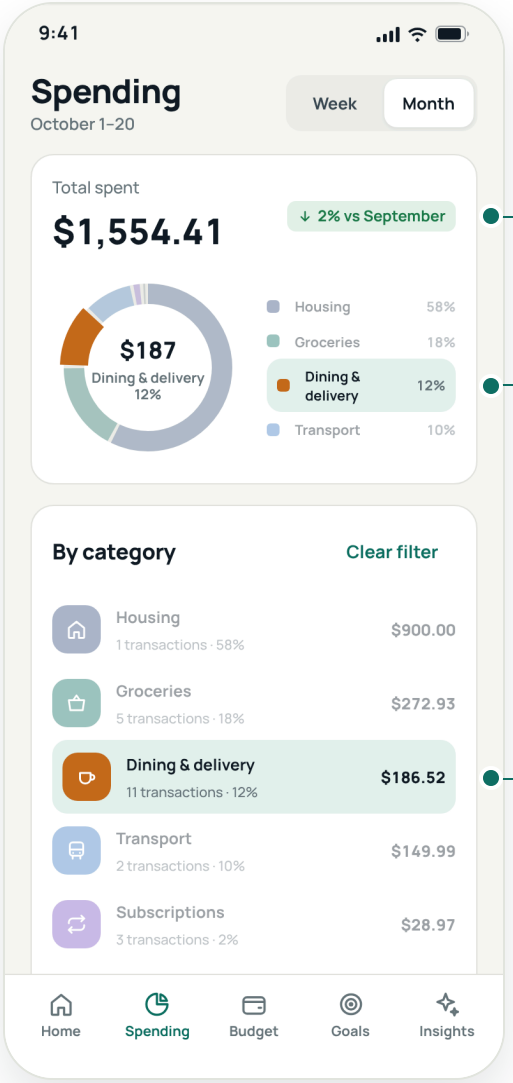
Dashboard · frame 04

1 Labelled time horizon
Runs to the next expected income, not to month end.

2 Bills and goals first
Income received, minus spending, bills due before payday and goal savings.

3 Exclusions are stated
The $400 expected Oct 30 isn't counted until it arrives. "How this is calculated" itemises every line.

4 Unpaid bills, with dates
Repeating payments still due this month. Those due before payday are marked "Set aside".



Spending, category selected · frame 08

5 Context for the total
Spending so far, compared with the same days last month.

6 One tap filters everything
Choosing a category, in the chart or a list, updates the donut centre, the daily bars and the transactions.

7 Not colour alone
Other categories dim, every row keeps its name, count and share, and tapping again clears the filter.

From customer problem to business value, with a pilot measure for every link

CUSTOMER PROBLEM	FINPATH FEATURE	PILOT OUTCOME MEASURE	POTENTIAL BUSINESS VALUE
Don't know what's safe to spend. Only 37% of Gen Z feel in control of their money. FACT	Safe to spend that counts only money received and reserves bills and goal savings; itemised on request. IMPLEMENTED	Weekly use of the safe-to-spend view; self-reported confidence +0.5 after 60 days. PROPOSED	More frequent digital engagement. Could reduce overdraft or NSF events if pilot data confirms it. HYPOTHESIS
Don't know where to start. 20% say so. FACT	Budget limits pre-filled from last month, editable in one step. IMPLEMENTED	Setup completion 70%; first-session plan (≥1 budget and ≥1 goal) 40%. PROPOSED	An earlier primary banking relationship with clients at the start of their financial lives. HYPOTHESIS
Saving feels abstract. ASSUMPTION A-4	Named goals with a target month and monthly amount. IMPLEMENTED	Share of users contributing to a goal in month 2: baseline set in the pilot. PROPOSED	Savings deposit growth if goals were later linked to real accounts or Find & Save (outside MVP; needs compliance review). HYPOTHESIS
Generic advice; unclear data use. ASSUMPTIONS A-1, A-2	Insights that show their math; a switch for each data use. IMPLEMENTED (insights are rule-based)	Insight opt-in 75%; usefulness ≥ 3.8 of 5. The prototype already records "Useful? Yes/No". PROPOSED	Trust in digital guidance; evidence on whether ML insights are worth building over rules (A-6). HYPOTHESIS

North-star metric

Share of active users who stay within budget and contribute to a goal in the same month. Baseline set in the pilot.

Business model

Free feature inside the existing banking app; no subscription. Value comes from engagement, retention and trust, not direct revenue.

What is not estimated

No revenue, cost or ROI figures. They need finance and product-owner input and pilot data.

Facts: TD Bank Group / Edelman DXI (Oct 2025). Targets are starting points to agree with stakeholders, not results.

Trust is a feature, so it was built and tested like one

Key risks

RISK	MITIGATION
Safe to spend overstates what's available	Counts only income received; reserves bills and goal savings; states what's excluded (BR-1 v2)
Insights read as advice	Education wording; no products (BR-5); compliance review
Duplicates NOMI	Build on NOMI data; validate bets B1–B3 with RBC Mobile owners
Wrong categories mislead	Recategorising on roadmap; insights cite their data
Irregular income breaks budgets	Diary study; term-based budgets next

Privacy by design

- Each data use has its own switch and a one-line explanation.
- Personalised tips and analytics sharing default to off (BR-6). In the prototype these two switches are stored but have no effect.
- Turning insights off removes all analysis views.
- Delete all data, after confirmation, clears the device (BR-7).
- Synthetic data only; no bank connection exists.

Accessibility: what was tested

- Automated axe-core 4.10 scan with the WCAG 2.0–2.2 A and AA rule sets: 6 screens and 2 open sheets, 0 violations.
- Invalid fields are marked aria-invalid, linked to their message, and receive focus.
- Background content is inert while a sheet is open; Escape closes it.
- Larger text setting; status never shown by colour alone.
- All buttons and inputs are at least 24 px (WCAG 2.5.8 minimum).

Open: chips, toggles and small buttons are below NFR-1's 44 px target. Automated rules cover only part of WCAG, so this is not a conformance claim. Not yet tested with a screen reader or with users of assistive technology.

About "AI-assisted"

The prototype's insights and "Ask FinPath" run on fixed rules and keyword matching, labelled as demo logic wherever they appear. A real AI model would need model-risk review, explainability standards and bias testing before any client use; the rule-based version is the baseline to compare it against (assumption A-6).

Validate before building more

NOW · DONE

Concept and prototype

- Business analysis pack: BRD, 16 FRs, 8 NFRs, 15 user stories
- 22-frame screen inventory and 17 component specs
- Working prototype: 93 of 93 automated checks pass
- Figma handoff: component specs and connection map (no Figma file yet)

NEXT · 6–8 WEEKS

Validate the riskiest assumptions

- Check current RBC Mobile and NOMI features against page 4
- 8–10 problem interviews with students and new grads (A-1, A-5)
- Moderated usability test, including "Can I afford this?" on safe to spend (A-3, A-7)
- Privacy, accessibility and screen-reader reviews

LATER · IF VALIDATED

Pilot and extend

- Pilot with real categorised data and NOMI Forecast predictions
- Term-based budgets for irregular income
- Meet the 44 px touch-target goal
- Open banking accounts after phase one; test ML insights against rules

Quality assurance summary

93 automated checks, all passing: 89 drive the prototype in a headless browser, 4 call calculation functions directly. QA found and fixed 6 defects, 3 of them accessibility issues.

WORKS	Onboarding, safe to spend, budgets, goals, charts, search, settings, deletion, persistence
DEMO LOGIC	Insights and Ask FinPath (rules, not AI)
SIMULATED	Loading delay and connection error
CONCEPT	Reminders, personalised tips, analytics sharing, life-stage tailoring, account linking

What this project demonstrates

Framing a problem with cited evidence, comparing honestly against an existing product, separating facts from assumptions, writing testable rules, designing a full state model, building a working prototype, and verifying it against its own rules.